

A nonlinear dynamic model for the Aircraft Reactor Experiment: Experimental validation

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ABSTRACT

We propose a dynamical compartmental model of the molten salt reactor in the Aircraft Reactor Experiment, and we assess its accuracy based on steady state and transient experimental data. We use modified point reactor kinetics equations to describe the power generation in the reactor core, and we model the thermal dynamics of the eight counter-flow heat exchangers using two compartments for the primary and secondary fluids and one compartment for the tubes. The core is modeled similarly with an additional compartment for the moderator. For the steady state experiments, we compare model predictions to experimental data for reported values, empirical estimates, and least-squares estimates of the thermal contact conductances and reported values of the thermal reactivity feedback coefficients. For the transient experiments, we use least-squares estimates of the thermal contact conductances together with reported values, estimates based on neutron flux Monte Carlo simulations, and least-squares estimates of the feedback coefficients. The accuracy of the model is low when using reported values for both the steady state and transient experiments, and in contrast, it is high when the model is fitted directly to the data using least-squares estimates. Furthermore, the steady state is predicted with acceptable accuracy for empirical estimates of the thermal contact conductances and reported values of the feedback coefficients, and the transient is also predicted with sufficient accuracy for least-squares estimates of the thermal contact conductances and estimates of the feedback coefficients obtained with Monte Carlo particle transport simulation.

1. Introduction

Molten salt reactors (MSRs) are one of the most promising next-generation reactor concepts for commercial power production. A key difference from earlier types of reactors is that the fissile fuel is dissolved into a molten salt which circulates through the core and is cooled by a set of heat exchangers. MSRs offer several benefits over conventional reactors, including high thermo-electric efficiency, low-pressure operation, and robust passive safety features. Additionally, continuous fuel reprocessing allows for a wider range of fuel cycle options, e.g., in waste-burners and thorium breeder reactors (Gen IV International Forum, 2002). However, it is still imperative to ensure safe and efficient operation, which is also an essential part of the design and licensing of a reactor. Therefore, it is necessary to develop mathematical models that describe the transient behavior of MSRs. Such models can be used to inform the design, analyze operating points, and evaluate control strategies (Ho et al., 2024).

The transient behavior of an MSR is governed primarily by neutron kinetics and thermal hydraulics. These are coupled through heat generation from fission processes, thermal feedback in the fuel salt, and delayed neutron precursor transport due to fuel flow (Lamarsh and Baratta, 2001). Therefore, a high-fidelity model would include the neutron transport equation (Duderstadt and Hamilton, 1976), three-dimensional Navier–Stokes equations, and energy balance equations that account for conduction, convection, and radiation. Such a model would allow for detailed predictions of heat generation and flow, and it would account for material properties, phase changes, and turbulence. However, in practice, it is not computationally tractable to perform simulations with such a model. Therefore, it is necessary to approximate the relevant phenomena, e.g., by neglecting three-dimensional variations in both heat and fluid flow properties.

A variety of approaches to modeling MSRs based on neutronics and thermal hydraulics have been proposed in the literature (Wooten and

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Powers, 2018). Each approach is distinguished by the level of detail with which it approximates the relevant phenomena. Neutron dynamics are typically modeled using 1) neutron diffusion, i.e., a partial differential equation, 2) quasi-statics, where the spatial and temporal dynamics are only loosely coupled, or 3) point reactor kinetics, where only the temporal dynamics of the neutron flux are considered, and the spatial distribution is assumed to be stationary. Furthermore, it is necessary to account for neutron precursor drift. For MSRs, the fission products that decay and emit a neutron after a significant time period (on the order of seconds) are circulated outside of the reactor core together with the fuel. This external circulation introduces *time delays* in the dynamical model equations, which are challenging because they may significantly affect the dynamics and stability of the system. Finally, the equations describing neutron and precursor transport are coupled to a thermal-hydraulic model that describes the heat transfer in the core and the heat exchangers.

One modeling approach is to modify the point reactor kinetics equations to account for precursor transport and couple them with a lumped heat flow model and a constant temperature reactivity feedback coefficient. Such models were developed at the Oak Ridge National Laboratory (ORNL) in order to characterize the dynamics of the Molten Salt Reactor Experiment (MSRE), which was operational between 1965 and 1969 (Ball, 1963; Ball and Kerlin, 1965; Kerlin et al., 1971). Singh et al. (2018b) have reproduced the MSRE model and extended it to the Molten Salt Breeder Reactor (MSBR) concept (Singh et al., 2017, 2018a). Later, they used the model to investigate plant-level dynamics (Singh et al., 2020). The model disregards spatial aspects of the neutron dynamics and thermal hydraulics, which means that the associated simulation time is reasonable. The above models show fair agreement with the available experimental data. Therefore, it is relevant to investigate whether the same modeling principles can accurately describe data from other experiments, e.g., from the Aircraft Reactor Experiment (ARE). The ARE involved an MSR built and operated at ORNL as part of the Aircraft Nuclear Propulsion (ANP) program, and it preceded the MSRE by ten years. Although it was less comprehensive and contained design features that were not originally intended for deployment as an MSR (Cottrell et al., 1955), it is a valuable source of experimental data on both steady state and transient operation of MSRs because very few experiments have been conducted so far. Therefore, it is relevant to test the validity of the above modeling approach using data from the ARE.

In this paper, we propose a dynamical model of the ARE based on the same principles used by Singh et al. to model the MSRE (Singh et al., 2017). We combine a set of point reactor kinetics equations, modified to account for precursor transport due to fuel circulation, with heat generation within lumped (i.e., nodal) representations of the fuel, moderator and coolant masses. The nodes (or compartments) are coupled to each other through equations describing convective and advective heat transfer. Furthermore, we account for time delays due to the circulation of the molten salt. Specifically, we represent the time delays explicitly in the differential equations. This results in a system of delay differential equations (DDEs), which we simulate using a numerical simulation software implemented in Python.¹ First, we assess the accuracy of the model's prediction of power generation, temperature differences, and heat removal in steady state operation, and we compare reported values (Cottrell et al., 1955) with empirical and least-squares estimates of thermal contact conductances (the remaining parameters are the same). The highest accuracy is obtained with the least-squares estimates, and we use them to assess the accuracy of the model's prediction of transient increases in power production during the removal and reinsertion of a control rod. We compare reported values of the thermal reactivity feedback coefficients with

estimates obtained from 1) Monte Carlo simulation and 2) least-squares estimation.

The remainder of this paper is structured as follows. Section 2 describes the ARE system and the modified point reactor kinetics equations, the heat transfer equations, and the power generation model. Section 3 describes how these equations are used to create the nodes of the lumped model and how these can be further combined to represent the subsystems that constitute the proposed mathematical model of the ARE system. Section 4 describes the methods used to estimate selected parameters, and Section 5 presents the results, i.e., the comparison between model predictions and the experimental data. Finally, conclusions are given in Section 6.

2. Theory and background

In this section, we present the background of the ARE and the theory of the model phenomena described in Section 3. The two phenomena are coupled through thermal reactivity feedback and external circulation of the fuel (including the neutron precursors) through heat exchangers.

2.1. The Aircraft Reactor Experiment

The ANP program was initiated at ORNL by the US Atomic Energy Commission (USAEC) for the purpose of developing nuclear propulsion for aircraft. The design of a nuclear powered jet engine is similar to that of a conventional jet engine. The main difference is that the compressed air is heated using heat from the reactor, rather than combustion heat. Therefore, there is no need for jet fuel and combustion chambers. The range of such engines would be limited only by the specific power of the fuel, which is orders of magnitude larger than that of jet fuel. Around 1950, the ANP decided to direct their efforts towards the construction and operation of an aircraft reactor. These efforts resulted in the ARE. The operating objectives for the experiment were to 1) attain high power density and fuel temperature and 2) gather data on the reactor's behavior and characteristics. The maximum sustained power reached 2.5 MW(th) at which the maximum fuel temperature was 860 °C with a 180 °C gradient across the core. Furthermore, during transients, the power reached up to 4 MW(th) with a maximum fuel temperature of 860 °C. The fuel was a NaF-ZrF₄-UF₄ (93.5% enriched) eutectic, and it circulated through a lattice of hexagonal beryllium-oxide blocks. The blocks served as both moderator and reflector, and they were cooled by a separate loop of liquid sodium. Both the fuel and sodium loops were cooled by helium loops, and heat was extracted from the helium with ambient coolant water and discarded (Cottrell et al., 1955). Fig. 1 shows a schematic of this configuration.

2.2. Point kinetics for molten salt reactors

The model proposed in this work describes the neutron dynamics with modified point reactor kinetics equations. Point reactor kinetics are a standard approximation in the field of nuclear engineering (Stacey, 2007; Duderstadt and Hamilton, 1976). However, it assumes that the fissile fuel is stationary. In MSRs, the fuel is circulated through heat exchangers outside of the reactor core. Therefore, in order to be applicable for MSRs, the description of the neutron precursor concentrations must be modified (Wooten and Powers, 2018). The resulting equations describe the concentrations of neutrons, n , and neutron precursors, C_i for $i = 1, \dots, N_g$, which are grouped into N_g groups depending on the decay rates:

$$\dot{n}(t) = \frac{\rho(t) - \beta}{\Lambda} n(t) + \sum_{i=1}^{N_g} \lambda_i C_i(t), \quad (1a)$$

$$\dot{C}_i(t) = \frac{\beta_i}{\Lambda} n(t) - \lambda_i C_i(t) - \frac{C_i(t)}{\tau_c} + \frac{C_i(t - \tau_l) e^{-\lambda_i \tau_l}}{\tau_c}. \quad (1b)$$

¹ <https://github.com/lukelabrie/msrdynamics>

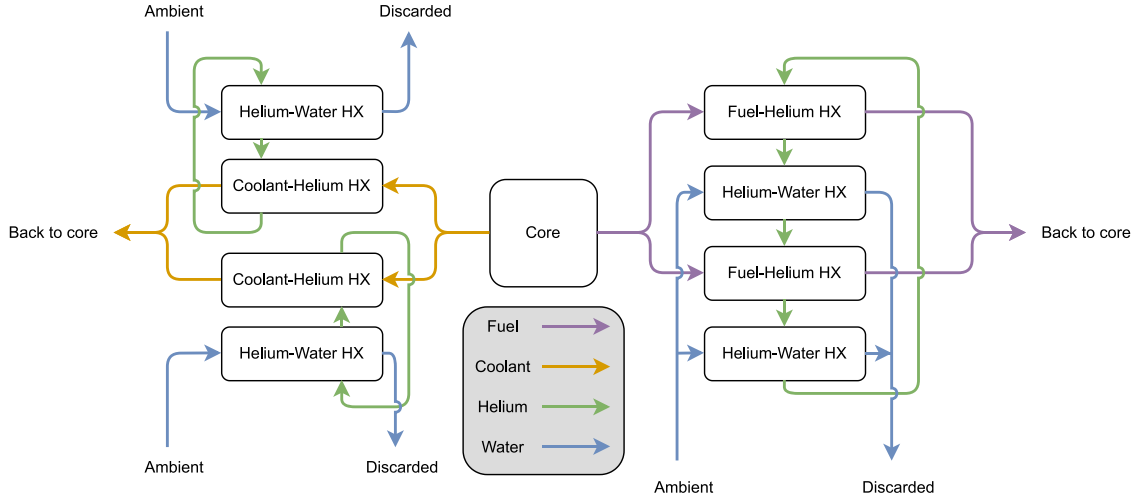


Fig. 1. A schematic of the reactor core and the heat exchangers in the ARE. Fuel and coolant are circulated through the core and cooled by helium through heat exchangers. Subsequently, the helium is cooled by water through another set of heat exchangers. Finally, the heated water is discarded.

The first term in (1a) describes the amount of neutrons generated by the fission events per unit time (per volume), and ρ is the reactivity, β is the delayed neutron fraction, and Λ is the mean generation time. The second term describes the amount of neutrons per unit time generated from the decay of neutron precursors. Furthermore, the first term in (1b) describes the amount of the i 'th neutron precursor group generated by the fission events, the second term represents the decay of the precursors, and the third term describes the loss of precursors due to the external circulation of the fuel per unit time. Here, β_i is the fractional yield for the i 'th precursor group, λ_i is the lumped decay rate of the i 'th precursor group, and τ_c is a time constant that depends on the ratio between the reactor volume and the volumetric flow rate (see also Appendix A), i.e., the transit time of the fuel through the core. Finally, the fourth term in (1b) describes the supply of neutron precursors from the external circulation stream, and τ_l is the time it takes the fuel to travel through the external loop. The reactivity, ρ , is a dimensionless quantity that characterizes deviations from criticality (Masterson, 2017). In the proposed model, it consists of a thermal feedback term, ρ_{fb} , and an external insertion term, ρ_{ext} :

$$\rho(t) = \rho_{fb}(t) + \rho_{ext}(t). \quad (2)$$

The change in the reactivity due to thermal feedback is affected by the change in the temperatures of the materials that contribute to the feedback effect, e.g., the fuel salt and the moderator:

$$\dot{\rho}_{fb}(t) = \sum_{i \in I} \frac{\alpha_i}{N_i} \dot{T}_i(t). \quad (3)$$

The set I contains the indices of the nodes in the model that contribute to the thermal feedback, and α_i is the thermal reactivity feedback coefficient of the material of the i 'th node. Furthermore, N_i is the number of nodes of a given material, e.g., fuel.

Remark 1. The modified point reactor kinetics equations are in the general form of mass balance equations describing chemical reactions taking place in a continuously stirred tank reactor (CSTR). Such equations have been studied extensively, and there exist many theoretical results, e.g., regarding stability. See Appendix A and, e.g., (Wahlgreen et al., 2020; Banaji et al., 2007) for further details.

2.3. Internal energy generation

The amount of energy generated in the fuel while it is circulating through the reactor core, Q_g , is directly proportional to the neutron concentration. It is given by

$$Q_g(t) = Q_{g,0} \frac{n(t)}{n_0}, \quad (4)$$

where $Q_{g,0}$ is the nominal power generated at the nominal neutron concentration, n_0 . The amount of energy deposited in the i 'th fuel node in the model is a fraction, k_i , of the total amount of power:

$$Q_{g,i}(t) = k_i Q_g(t). \quad (5)$$

2.4. Convective heat transfer

The heat transfer between a material at temperature T_j and a flowing fluid at temperature T_i in contact with the material, $Q_{c,i}$, is given by Newton's law of cooling, i.e.,

$$Q_{c,i}(t) = -k_{ij}(T_i(t) - T_j(t)), \quad (6)$$

where the thermal contact conductance k_{ij} is the product of the area of the interface between the material and the fluid, A_{ij} , and the convection heat-transfer coefficient, h_{ij} . The value of h_{ij} depends on flow conditions and the thermo-physical properties of both materials (Holman, 2010).

2.5. Advective heat transfer

The advective heat flow into the i 'th node is given by the difference between the enthalpies of the inlet and outlet stream (Holman, 2010, Chap. 5–10, Chap. 6–2). As the heat capacity, c_p , and the mass flow rate, f , are the same for the inlet and outlet streams (and because c_p is constant), the advective heat flow is given by

$$Q_{a,i}(t) = f(t)c_p(T_{in}(t) - T_i(t)), \quad (7)$$

where T_{in} is the temperature of the inlet stream and T_i is the temperature in the node (and in the outlet stream).

3. Mathematical model

In this section, we use the theory from Section 2 to formulate the proposed model of the ARE. First, in Section 3.1–3.5, we formulate the energy balances for four different types of nodes: The reactor fuel, tubes, coolant, and moderator in the reactor, and the primary and secondary fluids in the heat exchangers. Next, we describe the proposed model of the ARE in Section 3.6. The model consists of two types of subsystems, the reactor core and the heat exchangers, which are modeled as combinations of the individual nodes. In each subsystem, we model the solid materials with one node, and the fluid materials are modeled with two nodes. Finally, in Section 3.7, we describe the general form of the mathematical model equations. The reactor core

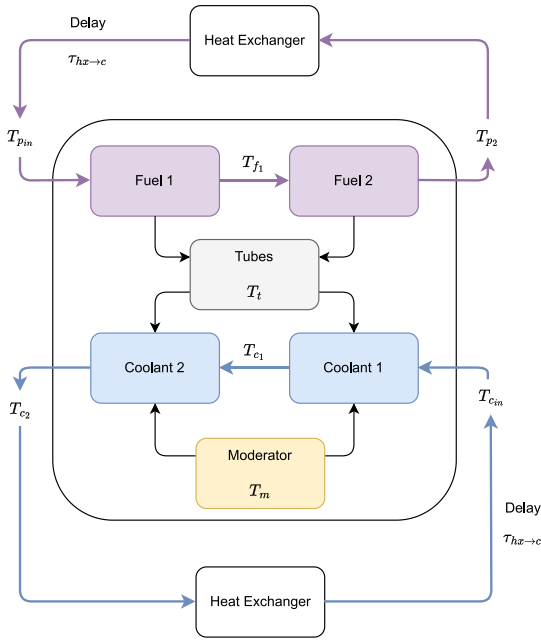


Fig. 2. Schematic of the model of the reactor core. The fuel and the coolant flowing through the reactor core are each represented by two nodes, and the moderator and the fuel tubing are each represented by one node. The fuel transfers heat to the fuel tubing, which transfers heat to the coolant whereas the moderator transfers heat to the coolant directly. The colored arrows indicate advective heat flows and the black arrows indicate convection.

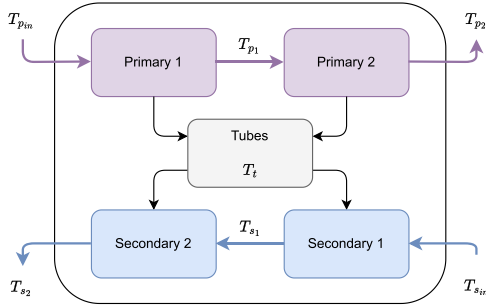


Fig. 3. Schematic of the models of the counter-flow heat exchangers. The primary and secondary fluids are each represented by two nodes, and the primary fluid transfers heat to the tubing, which transfers heat to the secondary fluid. As in Fig. 2, the colored arrows indicate advective heat flows and the black arrows indicate convection.

and the heat exchangers are sketched in Figs. 2 and 3, and the model of the ARE is sketched in Fig. 4.

Throughout this section, the notation $Q_{\gamma,i}^k$ refers to the heat flow from node i of type j (e.g., fuel) to a node of type k (e.g., tubes) due to phenomena γ (e.g., convection). The interpretation of $Q_{\gamma,i}^j$ is similar, and we omit the subscript i when there is only one node of type j in the subsystem. Furthermore, we comment on the signs of the heat flows under conventional operation of the reactor. However, in a given simulation, the signs may change due to changes in the operating point.

3.1. Core fuel node

The energy balance for a section of the fuel pipe in the reactor core (i.e., the i 'th node) is affected by internal generation, heat loss to the fuel tubes, and advection:

$$\dot{U}_i^f(t) = Q_{g,i}^f(t) + Q_{c,i}^{tf}(t) + Q_{a,i}^f(t), \quad i = 1, 2. \quad (8)$$

The amount of generated energy, $Q_{g,i}^f$, is always non-negative, and for the first fuel node (see Fig. 2), the inlet stream comes from a heat exchanger and the outlet stream is the inlet stream of the second fuel node. The outlet stream of the second fuel node is connected to a heat exchanger outside the reactor core. As the fuel is heated by fission processes in the reactor, the advection terms, $Q_{a,i}^f$, are typically negative. Furthermore, as the fuel is typically hotter than the tubes, the convection from the tubes to the fuel, $Q_{c,i}^{tf}$, will typically also be negative.

3.2. Tube node

We model the tubes in the heat exchangers and in the reactor core in the same way. In the core, the energy balance accounts for convection from the fuel and the coolant:

$$\dot{U}^t(t) = Q_c^{ft}(t) + Q_c^{ct}(t). \quad (9)$$

Typically, a positive amount of energy will be transferred from the fuel to the tubes, Q_c^{ft} , which will be transferred to the coolant, i.e., Q_c^{ct} is typically negative.

3.3. Core coolant node

The coolant in the reactor core removes energy from the tubes that are connected to the fuel and from the moderator. Furthermore, the coolant is circulated through the core. Consequently, the energy balance accounts for conduction and advection:

$$\dot{U}_i^c(t) = Q_{c,i}^{tc}(t) + Q_{c,i}^{mc}(t) + Q_{a,i}^c(t), \quad i = 1, 2. \quad (10)$$

As for the fuel nodes, the inlet stream of the first coolant node comes from a heat exchanger and the outlet stream is the inlet stream of the second coolant node. The outlet stream of the second coolant node is passed through the heat exchanger again. For the coolant nodes, the convection terms will typically be positive, and the advection terms are negative because the coolant nodes are heated as well.

3.4. Moderator node

The moderator in the ARE was solid beryllium-oxide blocks. Therefore, the corresponding energy balance does not include advective heat flow. However, there is a generation term due to neutron heating. The energy balance is

$$\dot{U}^m(t) = Q_g^m(t) + Q_c^{cm}(t), \quad (11)$$

where the convection from the coolant nodes to the moderator is given by $Q_c^{cm}(t) = -(Q_{c,1}^{mc}(t) + Q_{c,2}^{mc}(t))$, which is typically negative. Furthermore, the generation term, Q_g^m , is non-negative.

3.5. Heat exchanger fluid node

The primary (p) and secondary (s) fluid streams in the heat exchangers (see Fig. 3) are modeled identically. The energy balance of fluid j accounts for convection to the tubing and advection:

$$\dot{U}_i^j(t) = Q_{c,i}^{tj}(t) + Q_{a,i}^j(t), \quad i = 1, 2, \quad j \in \{p, s\}. \quad (12)$$

When the hot fuel serves as the primary fluid, we assume that no heat is generated in the heat exchanger. For the primary fluid (i.e., the hotter fluid), the convection term is

$$Q_c^{pt}(t) = -(Q_{c,1}^{tp}(t) + Q_{c,2}^{tp}(t)), \quad (13)$$

which will typically be negative, and the advection term, $Q_{a,i}^p$, will be positive. For the secondary (colder) fluid, the convection term is

$$Q_c^{st}(t) = -(Q_{c,1}^{ts}(t) + Q_{c,2}^{ts}(t)), \quad (14)$$

and the signs will be opposite.

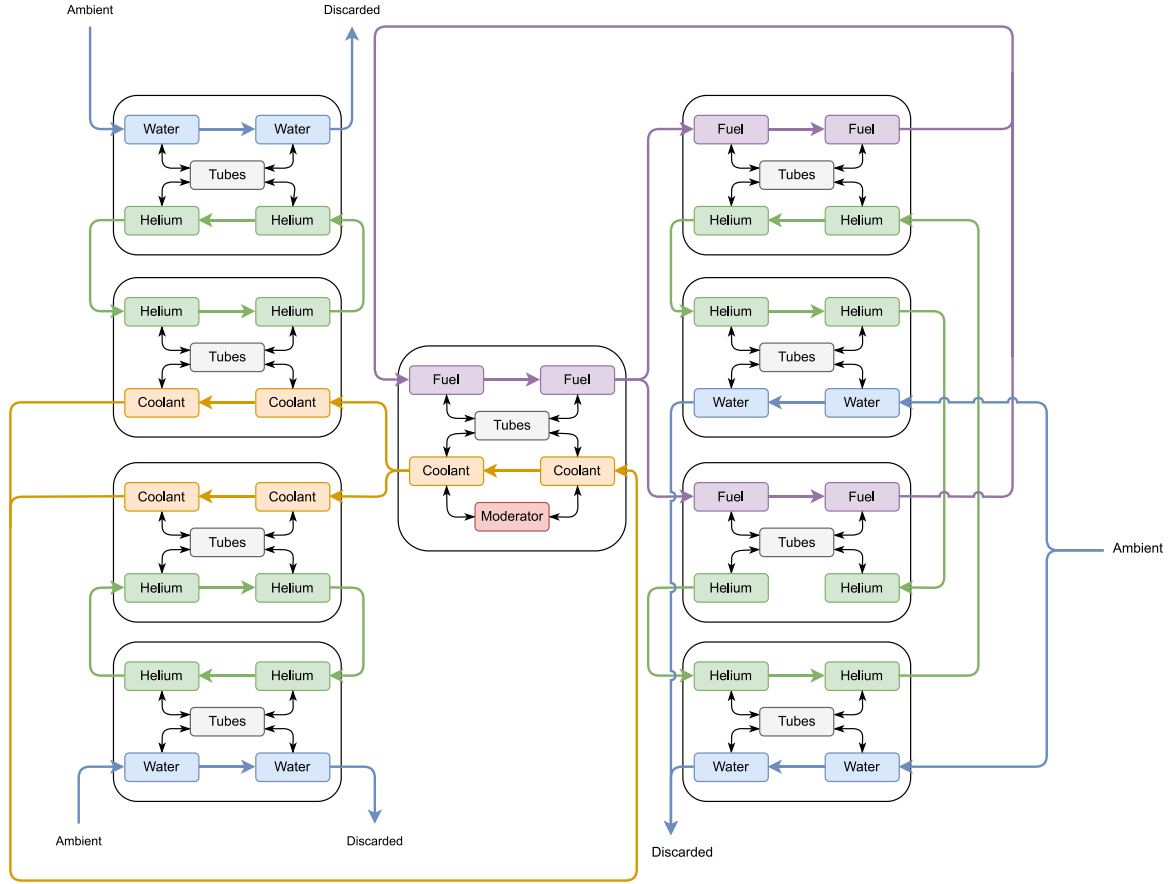


Fig. 4. Schematic of the model of the ARE reactor core and heat exchangers. The core and heat exchangers shown in Fig. 1 are modeled as shown in Figs. 2 and 3. Advective heat flows are indicated by colored arrows, and convective heat flows are indicated by black arrows.

3.6. Model of the aircraft reactor experiment

We model the reactor core in the ARE using the configuration shown in Fig. 2. As mentioned previously, the fuel and coolant pipes are modeled using two nodes whereas the tubing between them is modeled with one node. This choice is motivated by the work by Ball (1963) and balances model complexity with accurate representation of the dynamics of the system. The fuel and the coolant flow in opposite directions, and the moderator is bathed in the coolant. Both the fuel and the coolant are cooled by external heat exchangers, which introduces a time delay equal to half of the time delay in the point reactor kinetics equations for the neutron precursors (1b). In a counter-flow, shell-and-tube heat exchanger like those used in the ARE, the primary fluid flows through a lattice of tubes designed to maximize the heat transferred to the secondary fluid, which flows in the opposite direction on the other side of the tubes. Fig. 3 illustrates the chosen nodal configuration for the heat exchangers. We use these two subsystems to model the ARE system shown in Fig. 1. Fig. 4 shows an overview of the proposed model, which consists of the reactor core and eight heat exchangers that remove heat from the core. The heat from the fuel and the coolant is transferred to helium streams through four parallel heat exchangers (two for each), and the heat from the helium is transferred to water in four separate heat exchangers and discarded. Each coolant heat exchanger is cooled by a separate helium loop. Therefore, in principle, they do not need to be modeled separately. However, on the fuel-side, there is only one helium loop, which enters the fuel and helium heat exchangers in series. Therefore, the energy balances on the fuel-side require each heat exchanger to be expressed separately in the model.

The energy balances described in this section are in the form $\dot{U}_i^j(t) = Q_{i,tot}^j(t)$ where i is the compartment number, j is the fluid, and $Q_{i,tot}^j(t)$

is the sum of all heat flows affecting the compartment. We assume that the corresponding mass, m_i^j , and heat capacity, $c_{p,i}^j$, of the fluid are constant such that $\dot{U}_i^j(t) = m_i^j c_{p,i}^j \dot{T}_i^j(t)$, and in order to simplify the model equations, we rewrite each of the energy balances as

$$\dot{T}_i^j(t) = \frac{1}{m_i^j c_{p,i}^j} Q_{i,tot}^j(t), \quad (15)$$

such that the nodal temperatures are the model variables together with the concentrations of the neutrons and neutron precursor groups.

3.7. General mathematical form

The proposed mathematical model of the dynamics of the ARE consists of the modified point reactor kinetics equations (1), the reactivity equation (2), and the energy balances for each node in Fig. 4 reformulated as in (15). The model consists of differential equations, and there are time delays between some of the subsystems in the energy balances and in the modified point reactor kinetics equations. Furthermore, the modified point reactor kinetics equations are nonlinear whereas the reformulated energy balances are linear. Consequently, the model is in the general form

$$\dot{x}(t) = f(x(t), z(t), u(t), d(t), \theta), \quad (16a)$$

$$y(t_k) = g(x(t_k), \theta), \quad (16b)$$

where (16a) is a set of DDEs and (16b) is a set of algebraic measurement equations. Furthermore, $x(t) \in \mathbb{R}^{n_x}$ are the state variables, $z(t) \in \mathbb{R}^{n_z}$ are the memory states, $u(t) \in \mathbb{R}^{n_u}$ are manipulated inputs (e.g., external reactivity, ρ_{ext}), $d(t) \in \mathbb{R}^{n_d}$ are exogenous inputs (or disturbance

variables), $y(t_k) \in \mathbb{R}^{n_y}$ are the observed variables, and $\theta \in \mathbb{R}^{n_\theta}$ contains model parameters. The memory states are given by

$$z(t) = \begin{bmatrix} x(t - \tau_1) \\ \vdots \\ x(t - \tau_M) \end{bmatrix}, \quad (17)$$

where τ_i is the i 'th time delay, e.g., the transport time between the reactor core and an adjacent heat exchanger, and M is the number of time delays in the model. There are $M = 2$ time delays in the model. The first is $\tau_1 = \tau_l$ which occurs in the point reactor kinetics equations for the precursors (1b), and the second is $\tau_2 = \tau_l/2$ which occurs in the energy balances for the fuel and coolant in the reactor core and in the heat exchangers directly connected to the core. There are no time delays between any two heat exchangers.

3.7.1. Steady state

The steady state of the proposed model is not unique because the reactivity equation (2) is satisfied whenever the temperatures are constant, i.e., the steady state equations for the system (16a) are underdetermined. Therefore, we compute the steady state corresponding to a specified thermal energy generation in the reactor core.

First, we determine the steady state reactivity, $\rho_{fb,s}$, which is unique:

$$\rho_{fb,s} = \beta - \sum_{i=1}^{N_g} \beta_i \lambda_i \left(\lambda_i + \frac{1 - e^{-\lambda_i \tau_l}}{\tau_c} \right)^{-1}. \quad (18)$$

Next, we choose the steady state neutron concentration such that the total amount of heat generated in the reactor is equal to $Q_{g,s}$:

$$n_s = n_0 \frac{Q_{g,s}}{Q_{g,0}}. \quad (19)$$

Based on this value, we determine the steady state neutron precursor concentrations by

$$C_{i,s} = \left(\lambda_i + \frac{1 - e^{-\lambda_i \tau_l}}{\tau_c} \right)^{-1} \frac{\beta_i}{\Lambda} n_s. \quad (20)$$

Finally, given the total amount of heat generated in the reactor core, $Q_{g,s}$, and the inlet temperature of the water in the water-helium heat exchangers, the system defined by (16) is determined,² and the steady state temperatures in both the heat exchangers and the reactor core can be calculated using a symbolic solver. We use the symbolic Python toolbox SymPy (Meurer et al., 2017) to obtain the analytical expressions for these steady state temperatures.

4. Parameter estimation

The model of the ARE described in Section 3 contains a large number of parameters. Most of these parameters were measured and provided in the experimental reports. However, some were not reported. Additionally, the experimental uncertainties were not described. Therefore, in this section, we describe several methods for approximating the parameter values and present the different parameter estimates that are used in the steady state and transient analyses in Section 5. Furthermore, we discuss the different sources of uncertainty in the parameters and the experimental setup.

4.1. Empirical conductance estimates

For a counter-flow heat exchanger in steady state operation, an effective thermal contact conductance can be estimated from the temperature differences (Holman, 2010, p. 533). The estimate is given by

$$k_{ij} = -\frac{Q_{c,i,s}}{\Delta T_m}, \quad (21)$$

² In steady-state, delay terms simplify to $x(t - \tau_m) = x(t)$.

where $Q_{c,i,s}$ is the measured steady state heat transfer and ΔT_m is the log-mean temperature difference defined by

$$\Delta T_m = \frac{(T_{in}^p - T_{out}^s) - (T_{out}^p - T_{in}^s)}{\ln \left(\frac{T_{in}^p - T_{out}^s}{T_{out}^p - T_{in}^s} \right)}. \quad (22)$$

Here, T_{in}^p and T_{out}^p are the inlet and outlet temperatures of the primary fluid, and T_{in}^s and T_{out}^s are the inlet and outlet temperatures of the secondary fluid.

k_{ij} is also known as the overall heat transfer coefficient, which is often used to characterize how effectively a heat exchanger transfers heat between two fluids across all thermal resistances between them. To approximate the actual thermal contact conductances in the model, k_{ij} is applied to each node in proportion to its heat transfer area.

4.2. Least-squares estimation

For the transient analysis, we estimate the thermal feedback reactivity coefficients, α_i , in (3) based on measurements of the power, Q_g , in (4) from experiment H-8 (Cottrell et al., 1955). We denote the former (the parameters) by θ and the latter (the measurements) by y_k , and we estimate θ by solving a weighted least-squares dynamical optimization problem in the form

$$\min_{\theta} \frac{1}{2} \sum_{k=0}^N (y_k - y(t_k))^T W_k (y_k - y(t_k)), \quad (23a)$$

subject to

$$x(t) = x_s(\theta), \quad t \in [t_i, t_0], \quad (23b)$$

$$\dot{x}(t) = f(x(t), z(t), u(t), d(t), \theta), \quad t \in [t_0, t_f], \quad (23c)$$

$$z(t) = \begin{bmatrix} x(t - \tau_1) \\ \vdots \\ x(t - \tau_M) \end{bmatrix}, \quad t \in [t_0, t_f], \quad (23d)$$

$$y(t_k) = g(x(t_k), \theta), \quad k = 0, \dots, N, \quad (23e)$$

$$\theta_{\min} \leq \theta \leq \theta_{\max}, \quad (23f)$$

where $t_i = t_0 - \tau_{\max}$ is the initial time for the state trajectory, $\tau_{\max} = \max_{i \in \{1, \dots, M\}} \tau_i$ is the maximum time delay, and the weight matrix contains the squared inverses of the measurements on the diagonal:

$$W_k = \begin{bmatrix} y_{k,1}^{-2} & & & \\ & y_{k,2}^{-2} & & \\ & & \ddots & \\ & & & y_{k,n_y}^{-2} \end{bmatrix}. \quad (24)$$

The objective is to minimize the weighted squared deviation between the actual measurements, $\{y_k\}_{k=0}^N$, and the model predictions, $\{y(t_k)\}_{k=0}^N$, as described by the objective function in (23a). Furthermore, (23b) is an initial condition, (23c) are the dynamical constraints, and (23d) are the memory states. Finally, the model predictions of the observed variables are given by (23e), and (23f) are bounds on the estimated parameters. The system is assumed to be in steady state up to time t_0 , and the steady state is computed as described in Section 3.7.1. We approximate the solution to the dynamic optimization problem (23) using a shooting approach that transforms the infinite-dimensional problem into a finite-dimensional nonlinear program (NLP), which can be solved using off-the-shelf software. Specifically, we use the Python software JitCDDE (Ansmann, 2018) to approximate the solution to the initial value problem (23b)–(23d) for a given intermediate estimate of the parameters, θ , and we use SciPy's optimize (Virtanen et al., 2020) to approximate the solution to the NLP.

4.2.1. Steady state estimation

For the steady state analysis, we estimate the thermal contact conductances k_{ij} in (6) based on steady state measurements of the inlet and outlet temperatures of the fuel and coolant in the reactor core in experiment H-11 (Cottrell et al., 1955). We denote the former (the parameters) by θ and the latter (the measurements) by y , and in order to estimate θ , we solve a weighted least-squares problem in the form

$$\min_{\theta} \frac{1}{2} (y - y_s)^T W (y - y_s), \quad (25a)$$

subject to

$$y_s = g(x_s(\theta), \theta), \quad (25b)$$

$$\theta_{\min} \leq \theta \leq \theta_{\max}, \quad (25c)$$

where the steady state, x_s , is computed as described in Section 3.7.1, y_s are the steady state model predictions, y are the experimental measurements, and W is a weight matrix defined in the same way as in (24). Again, we use SciPy's optimize to approximate the solution to (25).

4.3. Monte Carlo particle transport

Experimental measurements of the thermal reactivity coefficients for the fuel, coolant, and moderator are available in the report by Cottrell et al. (1955). However, these can also be estimated computationally with a suitable particle transport software. Temperature-dependent density functions for each material are provided by Cottrell et al. (1955). These can be used to determine the density-related changes to the effective neutron multiplication factor (the ratio between the neutron production and loss rates). We use OpenMC to perform Monte Carlo simulations of the particle transport on a mesh generated from a publicly available CAD model of the ARE³ as shown in Fig. 5. The simulations are performed for N different temperatures, and OpenMC provides an estimate of the effective neutron multiplication factor, $k_{eff,j}(T_i)$, for each material j at temperature T_i together with a standard deviation. Next, we estimate the feedback coefficient of the j 'th material at the i 'th temperature, α_{ji} , by the temperature derivative of $\ln k_{eff,j}(T_i)$. For the lowest and highest temperatures, we use a forward and backward first-order difference approximation of the temperature derivative, respectively, and for the remaining temperatures, we use a central difference approximation. Finally, we estimate the reactivity of material j , α_j , by a weighted average of the corresponding estimates at the individual temperatures:

$$\alpha_j = \sum_{i=1}^N w_{ji} \alpha_{ji}, \quad w_{ji} = \frac{\sigma_{ji}^{-2}}{\sum_{i=1}^N \sigma_{ji}^{-2}}. \quad (26)$$

Here, σ_{ji} is the standard deviation corresponding to α_{ji} obtained from the standard deviations of $k_{eff,j}(T_i)$ through linearization of the natural logarithm, $\ln$.

4.4. Parameter sets

In Section 5, we compare experimental power and fuel temperature measurements with those predicted by the model. We consider both steady state and transient operation, and we assess the accuracy of the model for different parameter sets. The steady state power and temperature profile of the model is most sensitive to the thermal contact conductances (area times the convection heat transfer coefficients) in the core and heat exchangers. Therefore, we first estimate the conductances k_{ij} in (6) based on steady state measurements of the power, Q_g , in (4) and the inlet and outlet temperatures of the fuel and coolant from experiment H-11. For these estimates, we use the thermal reactivity

feedback coefficients reported by Lubarsky and Greenstreet (1953). In contrast, the transient behavior of the model is more sensitive to the thermal reactivity feedback coefficients, α_i , in (3). Therefore, for the transient analysis, we use the least-squares estimates of the thermal contact conductances and estimate the feedback coefficients based on 1) the results obtained with OpenMC and 2) the measurements of the power, Q_g , from experiment H-8.

Table 1 shows an overview of the parameter sets, and the values of the conductances and reactivity feedback coefficients are shown in Table 2. We use the reported parameters in S1 and T1 as baselines for assessing the accuracy obtained with the other parameter sets. We use S2 to assess the accuracy of using common empirical estimates of overall heat transfer coefficients to approximate the thermal contact conductances whereas for S3, we assess the model structure's ability to simultaneously capture the steady state of power generation, temperature drops across the core, and heat removal. The purpose of T2 is to evaluate the feasibility of predicting the transient response in power generation without using transient experimental data, and as for S3, we use T3 to determine whether the simple model structure can accurately represent the experimental data. Furthermore, for sets leading to high accuracy, the model can also be used in virtual experiments and to inform the design of model-based algorithms for, e.g., monitoring, prediction, control, and optimization. In general, we expect the least-squares estimates to most closely match the experimental data.

4.5. Sources of uncertainty

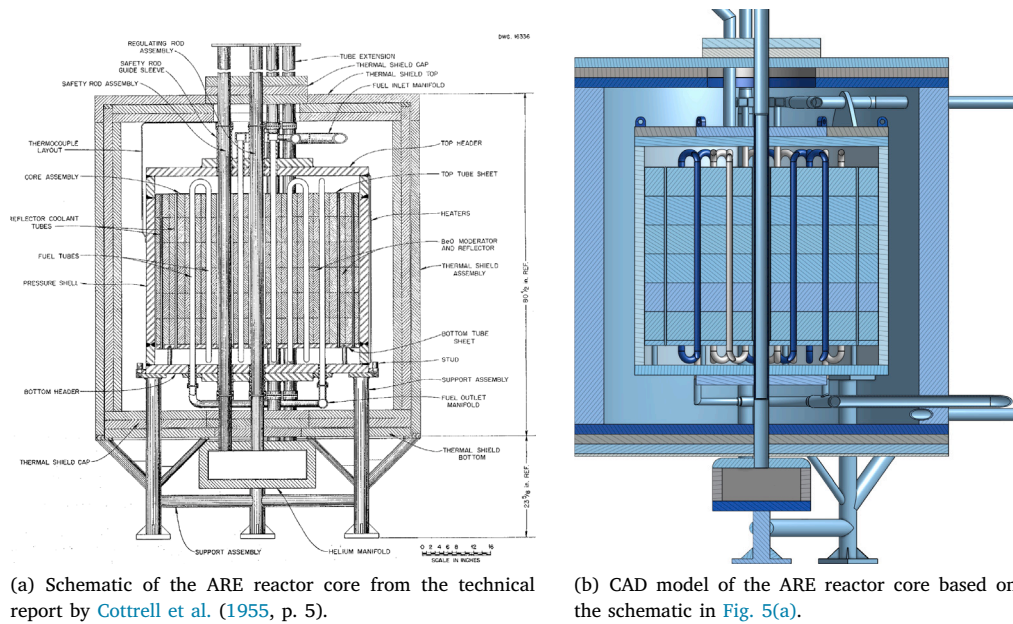
For a counter-flow heat exchanger, the inlet and outlet temperatures of both the primary and secondary fluids must be measured in order to estimate the thermal contact conductances. However, for all eight heat exchangers in the ARE, only the temperature of either the primary or secondary side of the loop was measured. For the temperatures that were not measured, we have used the design specification temperatures provided by Cottrell et al. (1955) instead. The differences between the design specification and measured temperatures are on the order of 10%. This potential discrepancy introduces significant uncertainty in the empirical estimates of the thermal contact conductances.

The thermal reactivity feedback coefficients were measured in experiment H-4, H-5, H-7, and H-10 (Cottrell et al., 1955), and the large differences between the values in the different sets in Table 2 may be due to severe issues with the temperature measurements in the ARE core. Consequently, any estimate based on core temperature measurements are inherently uncertain. Appendix K of the report by Cottrell et al. (1955, p. 180) describes that late in the experiment, the operators noticed that the thermocouples close to the core were consistently measuring significantly lower temperatures than those farther away from the core. This could be caused by radiation. The operators retroactively corrected the temperature data by correlating the erroneous temperature measurements with those that were assumed to be accurate (see also Appendix C). In this work, we have extracted the temperature measurements from the reactivity insertion experiment (H-8) from a strip chart recording. Consequently, it is possible that they were not corrected. This could explain the large differences in Table 2.

The experimental data for the transient assessment relies on time series data extracted from a tracing on page 79 of Cottrell et al. (1955). On the following page, a table provides the power level at four different stages of the experiment. At these stages, the discrepancy between the table data and the tracing ranges from 0% at the initial steady state, to 40% at the peak response to the reinsertion of the control rod. This suggests that either the data tracing or the data table is imprecise.⁴ Therefore, in Section 5.3, we compare the model predictions with both data sets (see Fig. 6).

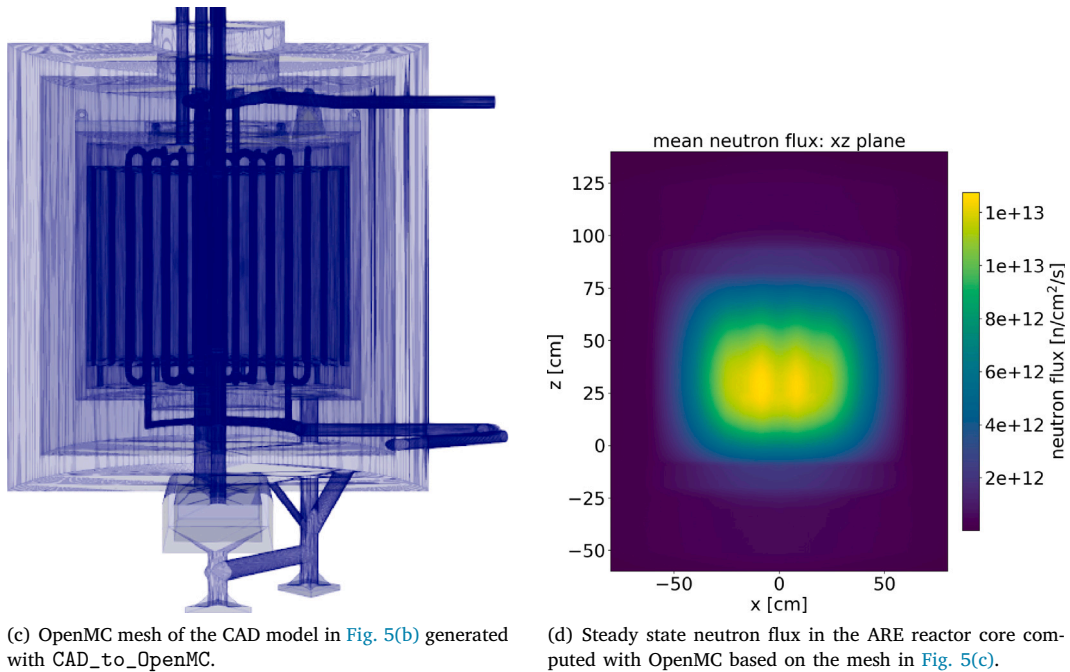
⁴ The data tracing appears to be hand-drawn and uses a y-axis with inconsistent intervals. Additionally, the data from the tracing is extracted using a digital tool. Therefore, we consider it more likely that the tracing itself is the source of the discrepancies, rather than the data table.

³ The CAD model of the ARE, along with several repositories related to MSR modeling, can be found at <https://github.com/openmsr>.



(a) Schematic of the ARE reactor core from the technical report by Cottrell et al. (1955, p. 5).

(b) CAD model of the ARE reactor core based on the schematic in Fig. 5(a).



(c) OpenMC mesh of the CAD model in Fig. 5(b) generated with CAD_to_OpenMC.

(d) Steady state neutron flux in the ARE reactor core computed with OpenMC based on the mesh in Fig. 5(c).

Fig. 5. A computer model of the neutron flux in the ARE reactor core. We compute the neutron flux with OpenMC based on a mesh generated from a CAD model of a schematic of the reactor core from the technical report by Cottrell et al. (1955). The OpenMC model uses 200 batches (out of which 10 are inactive) of 210000 particles.

Table 1

Overview of the parameter sets used in the steady state analysis (S1–S3) and in the transient analysis (T1–T3). Parameter sets S2 and S3 use the same thermal reactivity feedback coefficients as S1, and parameter sets T2 and T3 use the same thermal contact conductance estimates as S3. Furthermore, S3 and T1 are identical. The estimated parameter values are reported in Table 2, and the remaining model parameters are reported in Table B.4.

Name	Thermal contact conductances	Feedback coefficients	Section	Reference
S1	Reported values	Reported values	–	Lubarsky and Greenstreet (1953)
S2	Empirical estimates	Reported values (S1)	Section 4.1	
S3/T1	Least-squares estimates	Reported values (S1)	Section 4.2.1	Cottrell et al. (1955)
T2	Least-squares estimates (T1)	Particle transport estimates	Section 4.3	Romano et al. (2015)
T3	Least-squares estimates (T1)	Least-squares estimates	Section 4.2	

Table 2

The parameter values used in the steady state and transient analyses in Section 5 (see also Table 1). The direction of the heat flows (under conventional steady state operation) is indicated by arrows. In the core, heat is transferred from the fuel to the tubes and from the tubes to the coolant. Heat is also transferred from the moderator to the coolant. In the heat exchangers, heat is transferred from the primary fluids to the tubes and from the tubes to the secondary fluids. The thermal reactivity feedback coefficients are estimated for the materials that contribute the most to the thermal feedback: The fuel, moderator, and coolant.

Thermal contact conductances [kW/°C]								
Unit	Materials			S1	S2	S3/T1	T2	T3
Core	Fuel	→	Tubes	48.9		358		
Core	Tubes	→	Coolant	14.6	S1*	3.68		
Core	Moderator	→	Coolant	0.92		5.79		
Fuel-helium heat exchanger	Fuel	→	Tubes	6.09	3.35	55.2		
Fuel-helium heat exchanger	Tubes	→	Helium	1.40	3.35	2.07		
Helium-water heat exchanger (fuel-side)	Helium	→	Tubes	4.73	11.9	37.8	S3	S3
Helium-water heat exchanger (fuel-side)	Tubes	→	Water	343	11.9	2080		
Coolant-helium heat exchanger	Coolant	→	Tubes	10.3	1.43	101		
Coolant-helium heat exchanger	Tubes	→	Helium	0.45	1.43	1.00		
Helium-water heat exchanger (coolant-side)	Helium	→	Tubes	0.48	1.90	14.7		
Helium-water heat exchanger (coolant-side)	Tubes	→	Water	89.3	1.90	390		
Thermal reactivity feedback coefficients [pcm/°C]								
Material				S1	S2	S3/T1	T2	T3
Fuel				-17.6			-5.5	-5.3
Moderator				1.9	S1	S1	-2.1	1.4
Coolant				-10.6			-0.064	2.6

*Conductances cannot be estimated empirically in the presence of internal power generation. Therefore, for the core coefficients, we use the values from set S1.

Table 3

Steady state analysis based on the parameter values in Tables 2 and B.4. The analysis compares the experimental and predicted values of the generated power and the temperature increase in the fuel and coolant when passing through the core. The absolute and relative amounts of heat removed by the fuel and the coolant are also compared. For each parameter set, the predicted value (Pred.) is provided together with the relative error (Err.) with respect to the experimental value.

Variable		Experiment	Model (set S1)		Model (set S2)		Model (set S3)	
			Pred.	Err.	Pred.	Err.	Pred.	Err.
Generated power		2.10 MW	2.10 MW	–	2.10 MW	–	2.10 MW	–
Temperature increase (fuel)		174 °C	154 °C	–12%	181 °C	4.1%	175 °C	0.3%
Temperature increase (coolant)		61 °C	81 °C	33%	55 °C	–9.8%	61 °C	0.3%
Heat removal (fuel)	Absolute	1.52 MW	1.34 MW	–12%	1.58 MW	4.3%	1.53 MW	0.4%
	Relative	72%	64%	–12%	75%	4.8%	73%	0.4%
Heat removal (coolant)	Absolute	0.58 MW	0.76 MW	31%	0.52 MW	–11%	0.57 MW	–1.1%
	Relative	28%	36%	31%	25%	–12%	27%	–1.1%

5. Results

In this section, we assess the accuracy of the proposed model for the different parameter sets described in Section 4, and we consider both steady state and transient operation. In the steady state assessment, we compare with the measured thermal energy generation and the measured temperature difference across the reactor core, and in the transient assessment, we compare with the measured response in the power generation.

5.1. Parameter values

The estimates of the thermal contact conductances in S2 are consistent with the fact that the heat transfer between fluids is dominated by the fluid with the highest thermal resistance, i.e., the lowest thermal contact conductance. Overall, the estimates in both S2 and S3 reflect a lower overall thermal resistance between the core and the external coolant water than those of S1. The estimated values of the thermal reactivity feedback coefficient for the fuel are significantly lower in T2 and T3 than in T1. However, the sign is consistent, and the values for T2 and T3 agree to within 5% of each other. In contrast, the signs are not consistent across the parameter sets for the moderator and the coolant. For T2, the estimate is negative for the moderator and for T3, it is positive. Similarly, T2 estimates a comparatively small and negative coefficient for the coolant, while the estimate is comparatively large and positive in T3. The opposite signs between T2 and T3 might suggest

a change in the dominant physical phenomena, e.g., absorption relative to moderation. However, the results of the transient analysis are not as sensitive to changes in the moderator and coolant coefficients as they are to changes in the fuel coefficient, as can be seen in the results presented in Section 5.3. The coefficients in T3 agree more closely with the experimental data, but they also present a risk of over-fitting as they are only providing moderate improvement in the experimental validation at the expense of significantly differing from the Monte Carlo particle transport estimates used in T2.

5.2. Steady state experiment

Table 3 shows the results of the steady state assessment, and we choose the steady state power generation to match the measured value of 2.10 MW for all three parameter sets (see Section 3.7.1). For parameter set S1, the model underestimates the fuel temperature increase across the core by around 21 °C and overestimates the coolant temperature increase by almost the same amount. The fuel and coolant heat removal are under- and overestimated similarly. The model is more accurate for parameter set S2. In particular, the model predicts the fuel and coolant temperature increases within 7 °C of the experimental values. In contrast to S1, the fuel temperature increase is overestimated and the coolant temperature increase is underestimated. Finally, when the parameters are estimated with the least-squares approach (set S3), the temperature increases in the fuel and coolant are predicted within 0.3% of the experimental values. In conclusion, the steady state temperature increases and amounts of removed heat can be well described

by the model, and the accuracy of the model predictions is acceptable for empirical estimates of the thermal contact conductances.

5.3. Transient experiment

The experimental report by Cottrell et al. (1955) describes a reactivity insertion experiment (H-8) where a regulation rod was withdrawn and subsequently reinserted. During the experiment, the transient response in the generated power and the difference in the fuel temperature across the core were measured. We model the withdrawal and reinsertion of the regulation rod using the external reactivity in (2). Specifically, we use the profile shown in the bottom of Fig. 6, where the full withdrawal of the regulating rod was estimated to be equivalent to a 400 pcm reactivity insertion at a rate of 11 pcm/s (Cottrell et al., 1955, p. 78).

Fig. 6 shows the experimental measurements of power from both the data tracings (red dashed line) and the summary table (brown points) provided on pages 79–80 of (Cottrell et al., 1955), together with the model predictions for each parameter set. We also compare the model predictions to those of the high-fidelity model proposed by Martin et al. (2025). As we have already assessed the accuracy of the steady state predictions, we evaluate the predictions of the transient response relative to the steady state. We show deviations in the generated power in the fuel from the values at the initial time. After the regulation rod is removed, the generated power increases by 1.7 MW before reaching a steady state of 0.7 MW above the initial level. When the rod is reinserted, the generated power decreases to 1.5 MW below the initial value before reaching a steady state slightly below the initial state.

For parameter set T1 (which is the same as S3), the model significantly underestimates the response in the generated power. This may suggest that the thermal reactivity feedback coefficients reported by Cottrell et al. (1955) are too high, as the power responses are too heavily damped. The model is significantly more accurate for parameter sets T2 and T3, and the accuracy is higher for T3. The model predicts the transient response for T3 very well whereas it is slightly underpredicted for T2. For both sets, the intermediate steady state power generation between 150 s and 230 s into the experiment is underpredicted compared to the experimental data from both the tracing and the table. However, the response to the reinsertion of the regulation rod is overpredicted for both T2 and T3 compared to the table data and underpredicted compared to the tracing data. As most of the power is deposited in the fuel (around 90%), it is mainly the corresponding thermal feedback coefficient that determines the response of the generated power. The feedback coefficients are significantly lower for set T2 and T3 than for set T1. Therefore, the responses are not as heavily damped. Finally, the accuracy of the proposed model is comparable to that of the model by Martin et al. (2025) for the initial increase in power production, but qualitatively, the decrease is captured more accurately for the proposed model. In all cases, the models fail to capture the largely linear decrease in power production, but the model by Martin et al. predicts a short intermediate increase before settling at a steady state production significantly above that of the data (from both the tracing and the table).

6. Conclusions

We present a dynamical compartmental model of the reactor in the ARE based on modified point reactor kinetics equations and energy balances. The model includes a reactor core, two fuel-helium heat exchangers, two coolant-helium heat exchangers, and four helium-water heat exchangers, and we compare model predictions based on various parameter sets to both steady state and transient experimental data. The parameter sets contain different values of the thermal contact conductances and the thermal reactivity feedback coefficients. For the steady state data, we use reported values of the feedback coefficients, and assess the accuracy of the model with reported, empirical estimates, and

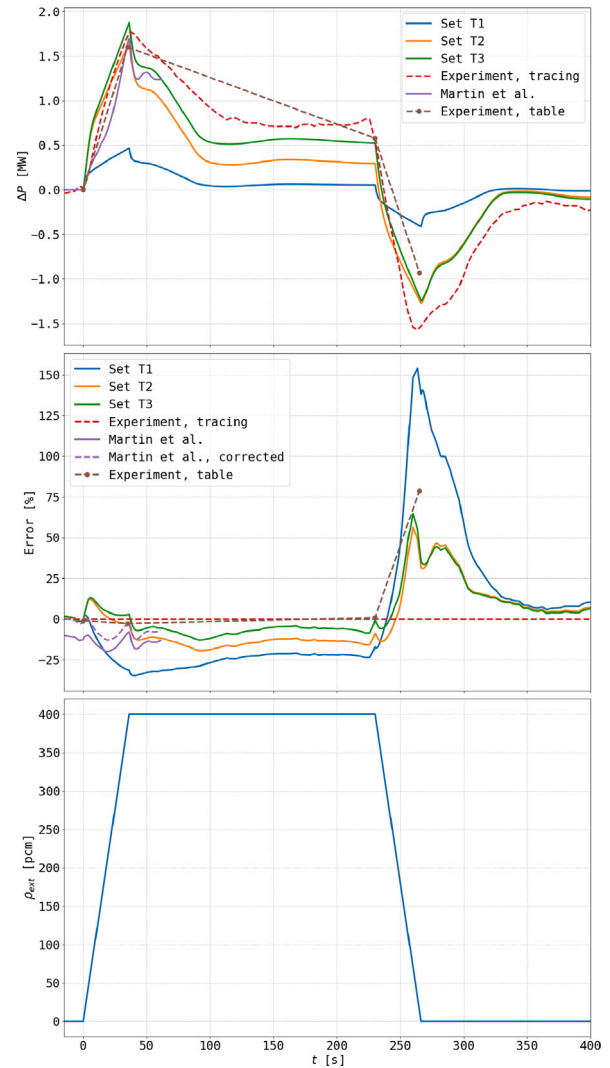


Fig. 6. Transient analysis based on the parameter values in Table 2 and B.4. The experiment involves the removal and reinsertion of a control rod (i.e., increase and subsequent decrease in the reactivity), and the experimental data (red dashed line and brown points) is compared with the model predictions based on parameter sets T1–T3 (blue, orange, and green lines, respectively), and the model prediction by Martin et al. (2025). Top: Deviation from steady state power generation, ΔP . Middle: Relative difference between model predictions and the experimental data from the tracing. The model proposed by Martin et al. (2025) assumes a steady state power generation of 2.1 MW for the transient experiment. However, the transient experiment was conducted with the reactor operating at 2.34 MW. To correct for this, the purple dashed line in the middle figure adjusts the power predictions by 0.24 MW before calculating the error. Bottom: The external reactivity insertion and withdrawal.

least-squares estimates of the conductances. The least-squares estimates provide the most accurate model in terms of temperature increases over the core and heat removed by the fuel and coolant. Therefore, for the transient data (reactivity insertion and removal), we use the least-squares estimates of the conductances and compare the model accuracy for reported, Monte Carlo particle transport estimates, and least-squares estimates of the thermal reactivity feedback coefficients. For the reported values, the deviation from the steady state power generation is significantly underestimated. For the Monte Carlo estimates and the least-squares estimates, the deviation in the generated power is predicted well during the reactivity insertion but is underestimated during the removal. In conclusion, the proposed model accurately

describes both steady state and transient power generation for least-squares estimates of the thermal contact conductances and thermal reactivity feedback coefficients, and the accuracy is also high when the feedback coefficients are estimated using Monte Carlo neutron flux steady state simulations.

A key advantage of the proposed model is that it is computationally tractable to, e.g., perform systems analyses, estimate the system state online, and inform the development of control algorithms. Additionally, the accuracy of the model may be improved by accounting for the uncertainty in parameters, inputs, and in the model structure, i.e., it could be extended to a stochastic model, which can also be used for uncertainty quantification (UQ) and robust decision-making and control. As the dynamical model focuses on predicting the transient response of the power generation to control rod insertion and withdrawal under normal or moderate deviations from normal operating conditions, future work could involve the design of optimal control strategies for ramping power generation, testing the robustness of existing control strategies, or informing the design of new molten salt reactors intended to operate dynamically. However, a key limitation of the model is that even after correcting for temperature measurement uncertainties (see Section 4.5), it does not predict the transient temperature drop across the core accurately (see Fig. 8 in Appendix C). Therefore, control or reactor design considerations that rely on temperature predictions should be based on an extension of the model or a high-fidelity model.

CRediT authorship contribution statement

Luke Labrie-Cleary: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Aslak Stubsgaard:** Conceptualization, Data curation, Formal analysis, Methodology, Supervision, Writing – review & editing. **John Bagterp Jørgensen:** Conceptualization, Formal analysis, Methodology, Supervision, Writing – review & editing. **Tobias K.S. Ritschel:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Luke Labrie-Cleary reports financial support was provided by Copenhagen Atomics A/S. Aslak Stubsgaard reports financial support was provided by Copenhagen Atomics A/S. They declare that this study received funding from Copenhagen Atomics A/S, which had the following involvement in the study: Provision of salary for the first author and second author and participation in the decision to publish. The other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. General form

The point-reactor kinetics equations (1) are a special case of the mass balance equations for a CSTR (Wahlgreen et al., 2020; Seborg et al., 2017) where a set of chemical reactions take place. Specifically, the model equations can be written in a more general form where the concentrations of the i 'th neutron precursor group and the neutrons are described by

$$\dot{C}_i(t) = (C_{i,in}(t) - C_i(t))D + R_i(t), \quad i = 1, \dots, N_g, \quad (27a)$$

$$\dot{C}_n(t) = R_n(t), \quad (27b)$$

respectively ($n = N_g + 1$). The dilution rate, $D = F/V = 1/\tau_c$, is the ratio between the inlet and outlet volumetric flow rate, F , and the volume, V , and τ_c is a time constant. Furthermore, the inlet concentration is

$$C_{i,in}(t) = C_i(t - \tau_l)e^{-\lambda_i \tau_l}, \quad (28)$$

and the production rate is

$$R(t) = S^T(t)r(t), \quad (29)$$

where the stoichiometric matrix and the reaction rates are

$$S(t) = \begin{bmatrix} -1 & & & 1 \\ & \ddots & & \vdots \\ & & -1 & 1 \\ \beta_1 & \dots & \beta_{N_g} & \rho(t) - \beta \end{bmatrix}, \quad r(t) = \begin{bmatrix} \lambda_1 C_1(t) \\ \vdots \\ \lambda_{N_g} C_{N_g}(t) \\ C_n(t)/\Lambda \end{bmatrix}. \quad (30)$$

Systems in the form (27) have been studied extensively (typically under the assumption that the stoichiometric matrix is constant in time), and there exist many theoretical results, e.g., regarding their stability (Banaji et al., 2007; Domijan and Kirkilionis, 2009).

Appendix B. Model parameters

Table B.4 shows the values of the parameters that are used across all of the parameter sets shown in Table 2 to compute the model predictions in Section 5.

Appendix C. Temperature corrections

As described in Section 4.5, the operators of the ARE noticed significant discrepancies in the temperature measurements obtained with thermocouples inside and outside of the core. It is likely that the discrepancy is caused by radiation inside the core. Cottrell et al. (1955) provided correlations of the temperature measurements in Figure K.2 in Appendix K of their report. These were used to estimate the actual temperature of the fluid in the core. However, we believe that this correction was not applied to the data tracings we have used to extract the experimental data from the reactivity insertion experiment. Furthermore, as the mathematical expressions for the correlations were not provided, we have used Web Plot Digitizer⁵ to obtain correlations between the temperature measurements inside and outside of the core. For the fuel inlet stream, the correlation between the inside, T_{in}^i , and outside, T_{out}^i , measurements is linear:

$$T_{in}^i = 0.90T_{out}^i + 68.45. \quad (31)$$

For the fuel outlet stream, the correlation between the inside, T_{in}^o , and outside, T_{out}^o , measurements is quadratic:

$$T_{in}^o = 0.002(T_{out}^o)^2 - 2.61T_{out}^o + 1526.10. \quad (32)$$

The temperatures in (31) and (32) are in °C, and Fig. 7 shows the correlations together with the data extracted from the report by Cottrell et al. (1955).

Fig. 8 shows the experimental measurements of the temperature difference across the core together with the model predictions corresponding to the analysis presented in Section 5.3 and shown in Fig. 6. Furthermore, it also shows the corrected experimental results obtained with the correlations (31)–(32). Although the model does not predict the times at which the temperature difference reaches its minimum and maximum deviation (both with and without corrections), the corrected experimental values are closer to the model predictions for parameter sets T2 and T3, which predicted the transient response in the generated power with decent accuracy.

⁵ <https://github.com/automeris-io/WebPlotDigitizer>

Table B.4

Model parameter values from the technical reports by Cottrell et al. (1955) and Lubarsky and Greenstreet (1953) used in the steady state and transient analyses presented in Section 5. The parameter sets S1–S3 and T1–T3 shown in Table 2 are used together with the parameter values in this table.

Parameter	Value	Unit	Source
Nominal power generation	2.10	MW	Cottrell et al. (1955, p. 185)
Delayed neutron fraction	470	pcm	Cottrell et al. (1955, p. 150)
Core fuel flow rate	2.901	L/s	Cottrell et al. (1955, p. 120)
Core coolant flow rate	9.588	L/s	Cottrell et al. (1955, p. 121)
Fuel density	2768	kg/m ³	Cottrell et al. (1955, p. 113)
Inconel density	8500	kg/m ³	Cottrell et al. (1955, p. 113)
Helium density	0.167	kg/m ³	Cottrell et al. (1955, p. 113)
Moderator density	2750	kg/m ³	Cottrell et al. (1955, p. 113)
Coolant density	0.1514	kg/m ³	Cottrell et al. (1955, p. 113)
Specific heat capacity of fuel salt	1.088	kJ/(kg K)	Cottrell et al. (1955, p. 113)
Specific heat capacity of Inconel 600	0.4237	kJ/(kg K)	Cottrell et al. (1955, p. 113)
Specific heat capacity of coolant	1.256	kJ/(kg K)	Cottrell et al. (1955, p. 113)
Specific heat capacity of helium	5.216	kJ/(kg K)	Cottrell et al. (1955, p. 113)
Specific heat capacity of moderator	2.014	kJ/(kg K)	Cottrell et al. (1955, p. 113)
Average fuel temperature	710.6	°C	Cottrell et al. (1955, p. 58)
Core fuel inlet initial temperature	653.9	°C	Cottrell et al. (1955, p. 120)
Core fuel outlet initial temperature	827.8	°C	Cottrell et al. (1955, p. 120)
Core coolant inlet initial temperature	663.3	°C	Cottrell et al. (1955, p. 121)
Core coolant outlet initial temperature	723.9	°C	Cottrell et al. (1955, p. 121)
Fuel loop transit time (τ_f)	47	s	Cottrell et al. (1955, p. 93)
Core transit time (τ_c)	8.3	s	Cottrell et al. (1955, p. 120)
Fraction of power deposited in fuel	89.2	%	OpenMC*
Fraction of power deposited in moderator	7.73	%	OpenMC*
Fraction of power deposited in Inconel 600	2.96	%	OpenMC*
Fraction of power deposited in coolant	0.147	%	OpenMC*
Volume of fuel in core	52.1	L	CAD Model
Volume of moderator in core	927	L	CAD Model
Volume of tubes in core	40.2	L	CAD Model
Volume of coolant in core	249	L	CAD Model
Volume of fuel in fuel-helium heat exchanger	23.5	L	Lubarsky and Greenstreet (1953, p. 47)
Volume of tubes in fuel-helium heat exchanger	5.62	L	Lubarsky and Greenstreet (1953, p. 47)
Volume of helium in fuel-helium heat exchanger	329	L	Lubarsky and Greenstreet (1953, p. 47)
Volume of coolant in coolant-helium heat exchanger	9.60	L	Lubarsky and Greenstreet (1953, p. 58)
Volume of tubes in coolant-helium heat exchanger	6.10	L	Lubarsky and Greenstreet (1953, p. 58)
Volume of helium in coolant-helium heat exchanger	59.5	L	Lubarsky and Greenstreet (1953, p. 58)
Volume of helium in helium-water heat exchanger (fuel side)	35.4	L	Lubarsky and Greenstreet (1953, p. 47)
Volume of tubes in helium-water heat exchanger (fuel side)	14.4	L	Lubarsky and Greenstreet (1953, p. 47)
Volume of water in helium-water heat exchanger (fuel side)	288	L	Lubarsky and Greenstreet (1953, p. 47)
Volume of helium in helium-water heat exchanger (coolant side)	10.9	L	Lubarsky and Greenstreet (1953, p. 58)
Volume of tubes in helium-water heat exchanger (coolant side)	4.45	L	Lubarsky and Greenstreet (1953, p. 58)
Volume of water in helium-water heat exchanger (coolant side)	65.1	L	Lubarsky and Greenstreet (1953, p. 58)

* Fractional heating values are estimated using OpenMC's `heatr` module with 200 batches (out of which 10 are inactive) of 210000 particles on the model shown in Fig. 5. Heating in materials other than fuel, moderator, and coolant is disregarded.

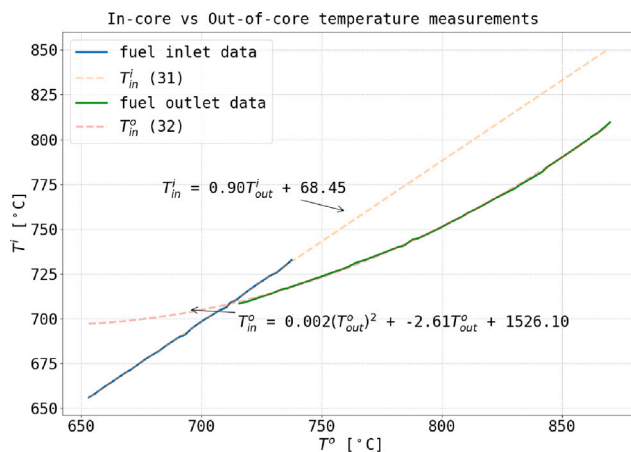


Fig. 7. Solid lines: Temperature correlation data extracted from Figure K.2 in Appendix K of the report by Cottrell et al. (1955) for the inlet (blue) and outlet (green) fuel streams. Dashed lines: The correlations in (31) and (32).

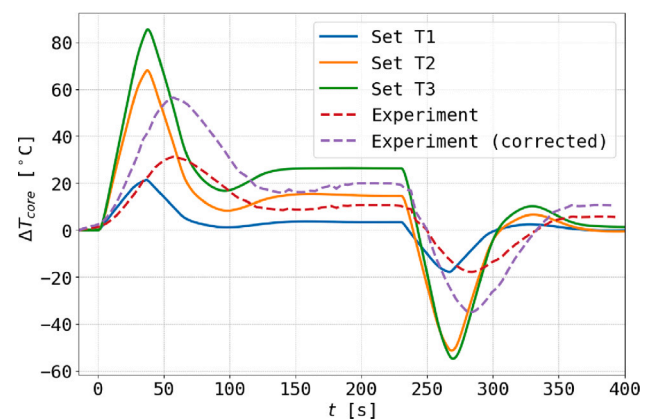


Fig. 8. Experimental measurements (with and without corrections) and model predictions of the deviation from steady state fuel temperature drop across the core, ΔT_{core} . The model predictions are obtained with the same model and parameter sets used to produce the results in the top of Fig. 6 (and the colors and line styles are consistent). The purple dashed line shows the corrected experimental data obtained with the correlations (31)–(32).

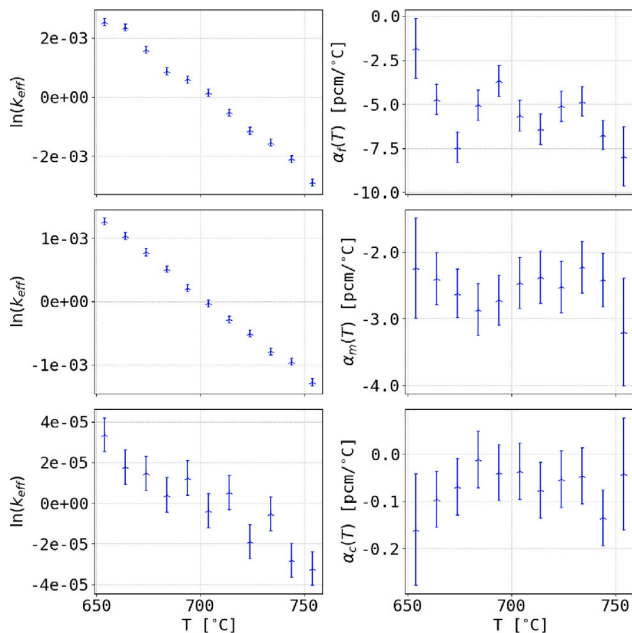


Fig. 9. Estimates of the natural logarithm of the effective neutron multiplication factor (left column) and finite difference approximations (see Section 4.3) of its temperature derivative (right column) for the fuel (top), moderator (middle), and coolant (bottom). The results are obtained from Monte Carlo simulations with the OpenMC software, and the error bars represent one standard deviation confidence intervals.

Table D.5

Parameters used in the OpenMC Monte Carlo simulations. The simulation results are used to estimate the thermal reactivity feedback coefficients for different materials.

Material	Particles	Batches	Inactive
Fuel	150,000	500	20
Moderator	2,000,000	200	20
Coolant	1,500,000	10,000	30

Appendix D. OpenMC Monte Carlo simulations

Section 4.3 describes how we use the Monte Carlo particle transport software OpenMC to estimate the thermal reactivity feedback coefficients for different materials. Fig. 9 shows the OpenMC Monte Carlo simulation results, and Table D.5 shows the parameters used to create the simulations. All simulations use the ENDF/B-VIII.0 nuclear cross section data library (Brown et al., 2018). The results (left column) exhibit a clear linear trend for the fuel and moderator (top and middle) whereas the trend is less clear for the coolant (bottom) because the order of magnitude is significantly lower. The left- and rightmost points in the right column have larger error bars (i.e., standard deviations) because they are obtained with a first-order finite difference approximation whereas the other points are obtained with a central difference approximation.

Data availability

The manuscript contains a link to a GitHub repository with code used to create the results presented in the manuscript.

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